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GA-SFS: A HYBRID ALGORITHM FOR FEATURE SELECTION IN BIOMEDICAL DATA

A thesis submitted in partial fulfillment of the requirements for the degree of Master of Science in Computer Science

By

Brian Bales

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The thesis of Brian Bales is approved:

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Vahe Karamian Date

—---------- —----------

Robert D McIlhenny Date

—---------- —----------

Dr. Rashida Hasan, Chair Date

California State University, Northridge

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ABSTRACT

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Brian Bales

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In biomedical data analysis, feature selection is crucial, particularly for high-dimensional datasets where redundant or irrelevant features might affect model performance. Biomedical datasets introduce additional challenges, such as high dimensionality, small sample sizes, and complex data structures which will then lead to overfitting, increased computation costs, and having trouble generalizing results. Addressing these issues often requires hybrid, adaptive, or ensemble-based algorithms that can improve model robustness and interpretability. To meet these demands, this paper introduces GA-SFS, a hybrid algorithm that combines Genetic Algorithms (GA) with Sequential Forward Selection (SFS) to select informative features in complex biomedical datasets effectively. GA explores the global search space to recognize relevant feature sets, while SFS refines the subsets by iteratively adding and removing features to have a more accurate classification performance. Experimental results on multiple datasets including real world biomedical applications, show that GA-SFS outperforms traditional feature selection methods in classification accuracy while reducing redundant or irrelevant features.

These findings highlight the potential of GA-SFS as a reliable and effective feature selection approach in biomedical research.

Index Terms—Feature Selection, Genetic Algorithm, Sequential Forward Selection, Biomedical Datasets, Hybrid Model

Chapter 1:

Introduction

This study will focus on improving feature selection in high dimensional biomedical datasets using a novel hybrid approach, GA-SFS, this approach combines Genetic Algorithms(GA) for global search optimization and Sequential Forward Selection(SFS) used for features subset refinement. Traditional feature selection methods often struggle with redundancy, computational inefficiency, and a decreased ability to capture complex feature relationships; this can negatively impact model performance. By integrating Ga and SFS, the method will enhance classification accuracy, reduce irrelevant features, and will improve computation costs. Thai research evaluates GA-SFS on various biomedical datasets, comparing its performance with existing feature selection techniques to demonstrate its effectiveness in biomedical applications.

1.1 Motivation

Machine learning and deep learning approaches have been applied widely in biomedical fields such as disease detection, medical imaging, genomics for identifying genetic markers, drug discovery for predicting compound efficacy, and personalized medicine for tailoring treatments [1], [4], [10]. Biomedical data encompass a wide range of information generated from biological and clinical sources, including genomic sequences, medical imaging, electronic health records, and wearable health sensors [6]. The datasets are often high-dimensional, with vast amounts of data points for each entry, and can contain features spanning genetic, molecular, and clinical domains [9]. In addition, these data often involve a large number of features collected from a relatively small number of samples [5], [12]. Many of these features may be irrelevant or

redundant, complicating the analysis and leading to potential issues such as overfitting [3], [7]. When non-informative features dominate, they can obscure the underlying patterns that are crucial for accurate predictions or classifications [2].

A primary challenge in biomedical data analysis is the lack of novel and adaptive feature selection algorithms able to handle the high-dimensional nature of such datasets [1], [13]. Traditional feature selection methods, such as filter-based or embedded model techniques, often struggle to find complex relationships among relevant features, which then leads to suboptimal model performance [14]. While wrapper-based methods, like Recursive Feature Elimination (RFE) and Genetic Algorithm (GA), often have improvements, they are computationally expensive and impractical for large-scale biomedical datasets [15]. Advancements in deep learning and hybrid optimization techniques present an opportunity to develop more robust feature selection methods that can better extract meaningful patterns from the data [8], [14]. Despite the growing interest in machine learning, there is still a gap in research when it comes to hybrid algorithms that balance efficiency, interpretability, and predictive accuracy in biomedical classification tasks [11].

1.2 Research Questions

Feature selection in machine learning is the process of selecting a subset of relevant features from a larger set to improve model performance while reducing computational complexity [4]. In biomedical data analysis, feature selection methods can be categorized into supervised, unsupervised, and semi-supervised approaches [5]. This study focuses on supervised feature selection, where features are selected based on their predictive relevance to the target variable.

Supervised feature selection methods fall into three main categories: filter, wrapper, and embedded methods [3]. Filter methods evaluate feature relevance independently using statistical tests or correlation measures, making them efficient for high-dimensional datasets but unable to capture complex feature interactions [6]. Wrapper methods, on the other hand, assess feature subsets using predictive models, often improving classification accuracy at the cost of increased computational expense [2], [10]. Embedded methods integrate feature selection into the learning process, offering a balance between efficiency and predictive power [12].

1. How can feature selection improve the performance of machine learning models in high-dimensional biomedical datasets?
2. What are the limitations of existing feature selection methods, and how do they impact biomedical classification tasks?
3. How can a hybrid approach combining Genetic Algorithms (GA) and Sequential Forward Selection (SFS) enhance feature selection efficiency and accuracy?
4. How does the proposed GA-SFS hybrid algorithm compare to traditional feature selection techniques in terms of classification performance and computational efficiency?
5. What impact does feature selection have on biomedical model performance when comparing models trained with and without feature selection?

1.3 Research Contribution

In this work, we propose GA-SFS, a hybrid feature selection algorithm that combines Genetic Algorithms (GA) and Sequential Forward Selection (SFS) to address the limitations of existing methods [15]. GA explores the feature space globally, identifying promising subsets through an

evolutionary optimization process, while SFS refines these subsets iteratively, ensuring improved classification accuracy and computational efficiency [16].

The key contributions of this research include:

- Development of a novel hybrid feature selection algorithm: GA-SFS combines the advantages of both filter and wrapper methods, leveraging GA for global search and SFS for local refinement [3]. This hybrid approach balances feature selection accuracy with computational efficiency.
- Comprehensive evaluation on multiple biomedical datasets: We conduct extensive experiments on diverse real-world biomedical datasets, demonstrating that GA-SFS consistently outperforms traditional feature selection methods in terms of classification accuracy and feature subset optimization [4], [17].
- Performance comparison with existing feature selection techniques: We systematically compare GA-SFS against state-of-the-art methods, highlighting its advantages in terms of model robustness, reduced feature redundancy, and improved interpretability [5], [18].
- Impact assessment of feature selection on biomedical model performance: We analyze and compare model performance with and without feature selection, demonstrating the importance of effective feature selection in biomedical applications [3], [7].

By addressing the challenges of high-dimensional biomedical data, GA-SFS contributes to improving disease classification, biomarker discovery, and other critical applications in biomedical research [4].

Chapter 2:

Related Work

In this chapter, we review existing research on feature selection methods in biomedical data analysis, focusing on both traditional and hybrid approaches [6]. This will highlight the strengths and limitations of existing techniques and establish the need for the proposed GA-SFS hybrid algorithm. Feature selection plays an important role in high-dimensional biomedical datasets, as redundant or irrelevant features can negatively impact model performance [3]. Traditional approaches, such as Genetic Algorithms (GA), Sequential Forward Selection (SFS), and Sequential Backward Selection (SBS), have been widely used but have certain limitations in terms of computational cost and handling feature interaction [12].

Newer and more recent approaches incorporate hybrid and adaptive techniques to enhance feature selection efficiency and accuracy [8].

2.1 Traditional Genetic Algorithm (GA) for Feature Selection

Genetic Algorithm (GA) is an effective optimization method inspired by the concepts of natural selection and evolutionary theory [24]. Research has shown its extensive applications in machine learning and artificial intelligence, particularly in feature selection [13], [15]. GA efficiently navigates large and complex feature spaces to identify the most relevant attributes for accurate label prediction [10]. Compared to other search techniques, GA emulates evolutionary mechanisms, such as selection, crossover, and mutation, to progressively refine subsets of features and enhance model efficacy [12], [16].

This makes GA advantageous in high-dimensional biomedical datasets, where irrelevant or redundant features can compromise classification accuracy, elevate computational cost, and

contribute to overfitting [14]. By leveraging GA's capabilities for global search, researchers can optimize feature selection in biomedical contexts, including disease classification, biomarker identification, and medical diagnosis [17], [19].

Process of Traditional GA:

1. **Initialization –** Define the initial population of feature subsets randomly.
2. **Fitness Evaluation –** Explain how subsets are evaluated using a classification model (e.g., SVM, KNN) and a performance metric (e.g., accuracy, F1 score).
3. **Selection Mechanism –** Discuss selection techniques like roulette wheel selection and tournament selection used to choose the best-performing feature subsets.
4. **Crossover and Mutation –** Explain how features are combined (crossover) and modified (mutation) to introduce diversity and avoid premature convergence.
5. **Termination Criteria –** The process stops when a defined number of generations is reached or when no further improvement is observed.

There are plenty of advantages for using Genetic Algorithm (GA) for feature selection, particularly in high dimensional biomedical datasets where traditional methods can struggle with scalability and feature dependencies. A key strength of GA is the ability to explore large feature spaces efficiently, making it effective for selecting the most relevant features. Unlike filter based methods that will evaluate individual features based on relevance independently, Ga will consider feature interactions, which will capture feature relationships and improve classification performance. Additionally, GA is more computationally feasible due to the use of evolutionary operators it will refine feature subsets without testing every possible combination.This capability makes GA more valuable in biomedical domains where samples will contain thousands of features but very little samples, such as some of the datasets used in this research.

2.2 Traditional Sequential Forward Selection (SFS) Methods

Sequential Forward Selection (SFS) is a method used for choosing the optimal features in a dataset [6]. It works by starting with an empty feature set and then gradually adding the most relevant features that enhance the accuracy of the classification model [12]. This process continues until adding more features no longer significantly improves performance or a predefined stopping criterion is reached [13]. SFS is particularly popular in biomedical research because it simplifies data while maintaining high predictive accuracy [5], [18].

Process of Traditional SFS:

1. Initialization – Start with an empty feature set.
2. Feature Selection – Iteratively add the feature that provides the highest performance improvement based on an evaluation metric (e.g., accuracy, F1 score, AUC-ROC).
3. Evaluation – Assess model performance using a classifier such as SVM, KNN, or Decision Trees.
4. Stopping Criterion – Stop adding features when the performance gain is negligible or a predefined limit (e.g., a maximum number of features) is reached.

Sequential Forward Selection (SFS) is a computationally efficient method for feature selection compared to other search techniques, SFS is used to incrementally select features rather than evaluate all the possible outcomes and combinations. This feature selection approach allows SFS to produce high performing feature subsets, which will create a better model generalizability while increasing the classification performance. Additionally, SFS is very easy to understand, making it a useful technique in biomedical datasets where model explanations are crucial in making clinical decisions. However, SFS is also known for its greedy approach which is a

notable drawback, a greedy approach means that once a feature is added, it cannot be removed, even if it later proves to be redundant. This will cause SFS to often fail to eliminate irrelevant or redundant features to a less accurate final prediction. Furthermore, SFS is also rather computationally expensive for large datasets while being significantly efficient for smaller datasets, making it less practical for high dimensional data like biomedical datasets.

2.3 Traditional Sequential Backward Selection (SBS) Methods

Sequential Backward Selection (SBS) is an alternative feature selection method that follows the opposite approach of SFS. Instead of starting with an empty set, SBS begins with all available features and progressively removes the least relevant ones based on a given performance metric [11]. This process continues until removing additional features significantly reduces model performance or a predefined number of features remain [21]. SBS is particularly valuable in biomedical research as it eliminates redundant or irrelevant features, leading to more efficient and interpretable models [12], [15].

Traditional SBS Process

1. Initialization – Start with the full feature set.
2. Feature Elimination – Iteratively remove the feature that has the least impact on model performance.
3. Evaluation – Assess model performance after each feature removal using a classifier like SVM, KNN, or Decision Trees.
4. Stopping Criterion – Stop when removing further features significantly reduces classification accuracy or another threshold is met.

Sequential Backward Selection (SBS) is an effective method for feature selection, especially for datasets that contain a large number of redundant or irrelevant features. SBS will incrementally remove the least significant feature instead of evaluating all possible combinations, leading to a more compact and efficient feature set. This will allow SBS to potentially have a much better outcome than Sequential Forward Selection (SFS) since it takes feature dependencies into account while removing attributes. Additionally, SBS will also increase the model interpretability, making it easier to determine which features are more important for the final prediction, which is valued in biomedical applications where understanding feature importance is crucial. SBS comes with its own challenges due to its high computational cost. Since it requires multiple iterations of model training and evaluation, it can get expensive for large datasets. Similar to SFS, SBS also has a notable limitation which is the risk of removing critical features too early leading to a decreased classification accuracy.

2.4 Deep Learning-Based Feature Selection Methods

Deep learning-based feature selection is a new idea that has emerged as a powerful approach for handling high-dimensional biomedical datasets. Methods such as autoencoders can be commonly used for the feature extraction and dimensionality reduction by obtaining compressed representations of the data while keeping important information through the process. Similarly, Deep Neural Networks (DNNs) inherently can perform feature selection by assigning different weights to features during the training, this will help to identify the most influential attributes. Lastly, reinforcement learning-based feature selection has gained attention for its adaptive ability to optimize feature subsets by dynamically selecting relevant features during training [14], [19].

While deep learning-based methods have potential in capturing complex, non-linear feature interactions, they also present challenges such as high computational cost, on large datasets. Many biomedical datasets have an issue with a small sample size, making deep learning based approaches tend to overfit. In contrast, hybrid methods like the proposed GA-SFS approach will help maintain a balance between global search, local refinement and redundancy all while ensuring interpretability and computation efficiency [12], [17], [23].

2.5 Research Gap and Contribution

Traditional methods like Genetic Algorithm (GA), Sequential Forward Selection (SFS), and Sequential Backward Selection (SBS) have notable limitations, despite their widespread use in biomedical feature selection [4]. These challenges prevent optimal feature selection and classification performance when applied independently. To address these issues, this study proposes the GA-SFS hybrid algorithm, which combines the strengths of each method while mitigating their individual weaknesses [23].

2.5.1 Limitations of Existing Methods

- Genetic Algorithm (GA) is effective for global search and optimization, efficiently identifying relevant feature subsets. However, it is computationally expensive and may suffer from premature convergence, leading to suboptimal feature selection [24].
- Sequential Forward Selection (SFS) ensures high classification accuracy by incrementally selecting the most relevant features. However, its greedy approach prevents removing redundant features, resulting in feature redundancy and inefficient models [9].

- Sequential Backward Selection (SBS) progressively eliminates irrelevant features, improving model efficiency. However, it is computationally expensive and risks removing important features too early, which can negatively impact predictive performance [11].

2.5.2 Addressing the Gaps

The proposed GA-SFS hybrid approach integrates the strengths of all three methods:

- GA for Global Search and Optimization – Ensures broad exploration of the feature space, preventing local optima traps and identifying promising feature subsets [24].
- SFS for Incremental Feature Selection – Works with GA to fine-tune feature subsets, selecting only the most relevant features while reducing redundancy [14].
- SBS for Feature Redundancy Reduction – Ensures that only essential features remain, improving model efficiency while preventing feature redundancy [18].

2.5.3 Contribution of This Research

This study introduces a novel hybrid feature selection approach that will promise the following for high dimensional biomedical datasets:

- An efficient hybrid feature selection method
    - The combined traditional methods will be used for its exploratory and refinement processes, ensuring a more optimal and balanced feature subset.
    - GA is being used to explore the global search space, while SFS and SBS will be used for the fine tuning of the selection process.
- Improved model performance with feature selection

    - The proposed method demonstrates a superior classification accuracy and F1 scores compared to the State-of-the-art feature selection techniques across multiple datasets.
    - Model performance remains high while also reducing the number of selected features, making it ideal for lower computation costs in applications such as medical.
- Robustness Against Overfitting
    - SFS which only adds features or SBS that only removes features, this approach balances the removal and addition of features, ensuring the final model is not overfitted to the training data.
- Practical Application to Biomedical Datasets
    - Applied on real world application of biomedical dataset such as cancer classification and gene expression analysis.

These three methods after being integrated will bridge the gap between global optimization and local refinement, creating an effective hybrid approach for feature selection framework for biomedical applications. The next chapter will get into a more detailed explanation of the implementation and evaluation process of the approach [23].

Chapter 3:

Proposed Methodology

In this chapter, we introduce GA-SFS, a hybrid feature selection algorithm designed to improve the feature selection process in high dimensional biomedical datasets. Traditional feature selection methods often struggle with redundancy and lack the ability to capture complex feature interactions. To address these challenges the following method that combines Genetic Algorithms (GA) with Sequential Forward Selection (SFS) and Sequential Backward Selection (SBS), is being proposed to combat these challenges of the traditional methods. The architecture of our model consists of three key methods, the GA method, responsible for generating and evolving candidate feature sets. SFS and SBS used to refine the selection and remove redundancy in the feature set. Our approach is evaluated using Support Vector Machines (SVM) with five-fold stratified cross-validation to ensure robustness. The algorithm will integrate tournament selection, two point crossover, and adaptive mutation torefin feature selection iteratively. Finally the SFS and SBS will be applied to further refine the optimized feature selection. The section will go in detail about the structure of our hybrid model, and the step-by-step methodology used to implement GA-SFS.

3.1 Problem Statement

Biomedical datasets will often contain a high number of features with a low number of samples, leading to overfitting, increased computational complexity and reduced model interpretability. As mentioned, the traditional methods that are used in this research often have limitations when used independently, such as redundant feature selection, premature convergence, or the risk of removing important features. This hybrid approach that is proposed in this research optimally

selects relevant features while minimizing redundancy and improving classification accuracy in biomedical applications.

Definition 1: Biomedical Data: Biomedical data refers to high-dimensional information from various medical and biological sources, including genetic data, clinical records, and medical imaging. The term ”high-dimensional” is defined as datasets containing a large number of features (e.g., genes, biomarkers, or imaging attributes) in relation to the relatively small number of available samples.

Definition 2: Feature Selection in Supervised Learning: Given a dataset with a high-dimensional feature space, feature selection refers to the process of identifying and selecting a subset of the most relevant features from the larger set to improve model performance and reduce computational complexity. The problem of feature selection in supervised learning can be formally defined as:

$$S^* = \arg\max_{S \subseteq F} A(S) \qquad \text{Eq(1)}$$

Where: S^* is the optimal subset of selected features, S is a candidate subset of features, F is the set of all available features, and $A(S)$ is a performance metric (such as accuracy, F1 score, or cross-validation error) evaluated on the selected feature subset S.

3.2 Proposed Model Algorithm

Our hybrid approach consists of three modules: The genetic algorithm module to choose the candidate feature set, the Sequential forward selection module, and the Sequential backward selection module to select the final optimal feature subset.

1) Genetic Algorithm: Our feature selection approach GA-SFS leverages a Genetic Algorithm (GA) to systematically find an optimal subset of features from high-dimensional biomedical data. This algorithm operates through the following sequence of steps:

• Initialization: We begin by randomly generating an initial population of chromosomes, each representing a unique combination of features. To encourage variety, each chromosome includes at least three features, covering a broad range of potential feature combinations from the start [2].

• Fitness Evaluation: We evaluated each chromosome's fitness using a Support Vector Machine (SVM) classifier with five-fold stratified cross-validation to ensure consistent class distribution across folds. The F1 score is chosen as the fitness metric, reflecting the need to balance precision and recall in biomedical datasets, which often face class imbalance challenges. Given a chromosome's feature subset S, the fitness $F(S)$ is evaluated by training an SVM model with five-fold Stratified K-Fold cross-validation. For each fold i, the F1 score is computed as:

$$F1_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \qquad \text{Eq(2)}$$

The overall fitness of the chromosome's feature subset S is the average F1 score across all folds:

$$Fitness(S) = \frac{1}{k}\sum_{i=1}^{k} F1_i \qquad \text{Eq(3)}$$

where k is the number of folds. This approach balances precision and recall across folds, ideal for managing class imbalances in biomedical datasets [8], [16].

• Selection (Tournament Selection): Chromosomes are selected as parents through a tournament selection strategy, where the fittest chromosome from a random subset is chosen. This selection process introduces selective pressure, increasing the likelihood of choosing chromosomes with higher fitness while maintaining diversity in the population. The process of tournament selection can be mathematically represented as:

$$Parent = \arg\max_{i \in T} f(x_i) \qquad \text{Eq(4)}$$

Where: T is a randomly chosen subset of individuals from the population, $f(xi)$ is the fitness function evaluated for individual i in the subset T , and arg max indicates selecting the individual with the highest fitness in the subset T as the parent [10].

• Crossover (Two-Point Crossover): Selected parent chromosomes undergo a two-point crossover, where two sections of the chromosomes are swapped. This allows off-spring chromosomes to explore new feature combinations by blending the best features from each parent. Consider two parent chromosomes $P1$ and $P2$, each consisting of a set of features. The two-point crossover involves selecting two crossover points, say at positions $cp1$ and $cp2$, where

$cp1 < cp2$, and swapping the feature subsets between these two positions. For each offspring Oi: Select two crossover points $cp1$ and $cp2$ randomly and create Oi by combining segments from two parents [7], [18]. The crossover operation can be mathematically represented as:

$$O_i[1 : cp_1] = \text{Parent1}[1 : cp_1] \quad \text{Eq(5)}$$

$$O_i[cp_1 : cp_2] = \text{Parent2}[cp_1 : cp_2] \quad \text{Eq(6)}$$

$$O_i[cp_2 : F] = \text{Parent1}[cp_2 : F] \quad \text{Eq(7)}$$

• Adaptive Mutation: To maintain diversity, we use an adaptive mutation rate, meaning chromosomes with lower fitness are given higher mutation rates. This encourages exploration in parts of the feature space that may have been overlooked, helping the algorithm avoid getting stuck in local optima. For each gene g in Oi, the adaptive mutation is defined as:

$$adaptive_rate = \mu \times \left(1 - \frac{\text{Fitness}(O_i)}{\max \text{Fitness}(\text{Pop}_g)}\right) \quad \text{Eq(8)}$$

where

– $popg$ represents the gene g in the population

2) SFS : The second component of our hybrid model is a combination of Sequential Forward Feature Selection (SFFS) and Sequential Backward Selection(SBFS) where we iteratively add and remove features to find the optimal subset [5], [14]. In this process, we use the AUC-ROC score to evaluate the performance of the selected feature subset. The AUC-ROC score serves as a robust measure of a model's capability to differentiate between positive and negative classes, making it highly applicable to biomedical datasets. Let F represent the set of all available

features, and let S^* denote the optimal subset of features. The goal is to iteratively select a subset of features S^* that maximizes the AUC-ROC score. The optimization function can be written as

$$S^* = \arg\max_{S \subseteq F} \text{AUC-ROC}(S) \qquad \text{Eq(9)}$$

Where: S^* is the optimal subset of features, AUC-ROC(S) is the AUC-ROC score calculated for the feature subset S, and F is the set of all available features. The process of SFFS with AUC-ROC is as follows:

• Initialization: Start with an empty set of selected features

$(S^* = \emptyset)$

• Iteration: At each step, add the feature $f_i \in S \setminus S^*$ that maximizes the AUC-ROC score:

$$S^*(t+1) = S^*(t) \cup \{f_i\},$$

$$f_i = \arg\max_{f \in S \setminus S^*(t)} \text{AUC-ROC}(S^*(t) \cup \{f\}) \qquad \text{Eq(10)}$$

• Selection: After adding a new feature, remove features that reduce the AUC-ROC score:

$$S^*(t+1) = S^*(t) \setminus \{f_j\},$$

$$f_j = \arg\min_{f \in S^*(t)} \text{AUC-ROC}(S^(t) \setminus \{f_j\}) \qquad \text{Eq(11)}$$

3) Hybrid Module: Our hybrid module combines the strengths of Genetic Algorithms (GA) and Sequential Forward Floating Selection (SFFS) to optimize feature selection. Initially, the GA is

employed to explore the feature space, using evolutionary processes such as selection, crossover, and mutation to generate diverse feature subsets. Once the GA has identified potential subsets, the process is refined by SFFS, which iteratively adds or removes features based on their impact on the AUC-ROC score. The general framework of our approach is shown in Fig1

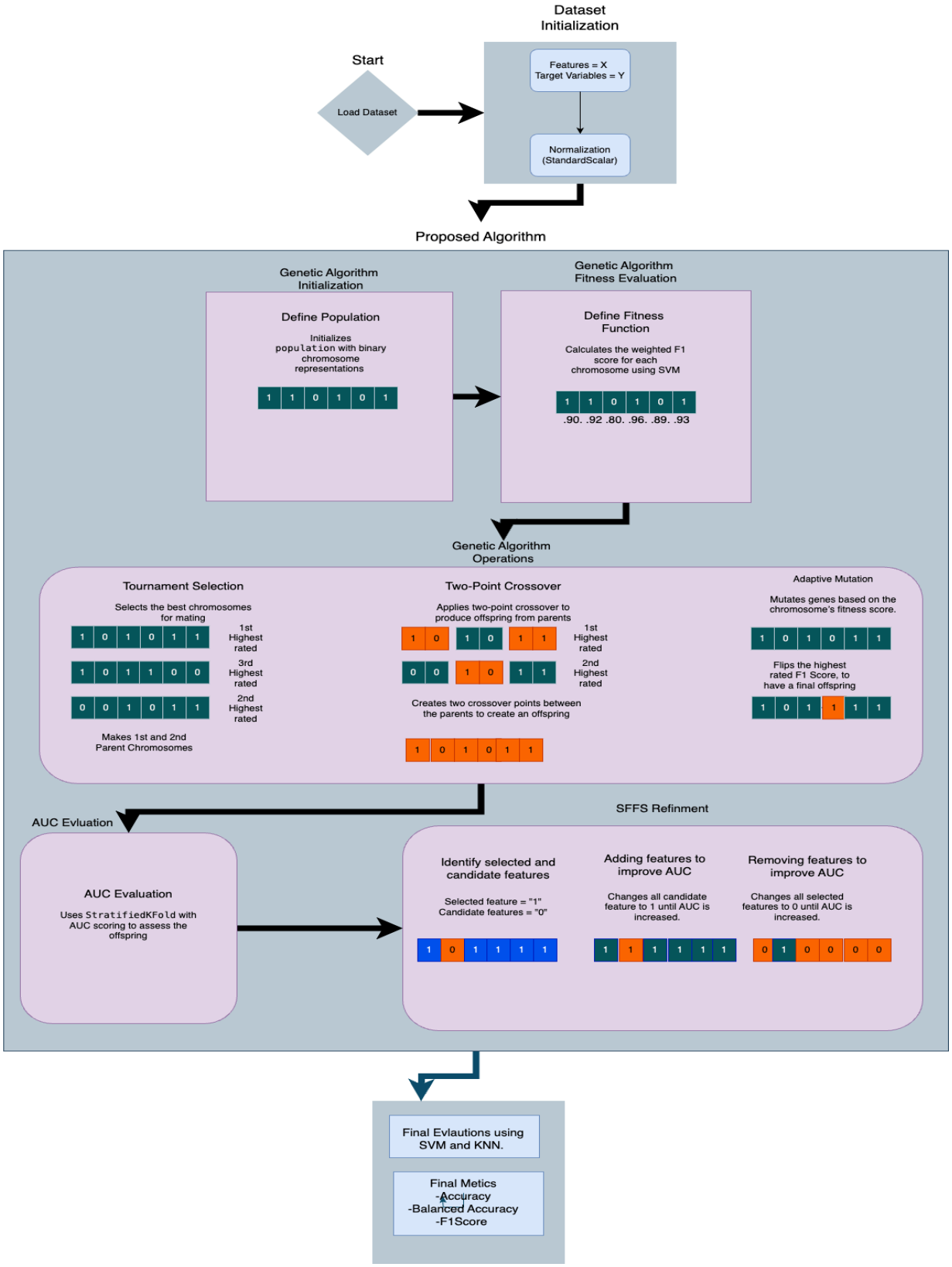

Fig. 1: The General Framework of GA-SFS

Chapter 4:

Experiments and Results

This chapter will present the experimental setup and results obtained from using GA-SFS hybrid feature selection algorithm on multiple biomedical datasets. The objective is to be able to demonstrate the effectiveness of GA-SFS in selecting the most informative features while improving classification accuracy and computation efficiency. The hybrid approach will be conducted on six publicly available biomedical datasheets: WDBC (Wisconsin Diagnostic Breast Cancer), WPBC (Wisconsin Prognostic Breast Cancer), SPECT (Single Proton Emission Computed Tomography), SBRCT (Small Round Blue Cell Tumors), Prostate, and Colon. Section 4.1 provides an overview of these datasets, including their characteristics and class distributions. Section 4.2 will detail the experimental setup, including parameter tuning, classifier selection (KNN and SVM), and the use of five fold stratified cross-validation to ensure reliable evaluation. In section 4.3, we compare mode performance with and without feature selection, highlighting the impact of GA-SFS on classification accuracy. Section 4.4 presents a comparative analysis between GA-SFS and the State-of-the-art feature selection methods, which will demonstrate the superior performance across different evaluation metrics. Finally, Section 4.5 discusses the computational efficiency and real world applicability of GA-SFS, specifically when trying to detect cancer in the early stages and biomarker discovery, where selecting relevant features is crucial for improving the decision making of the model.

4.1 Datasets Description

We collect the dataset from the UCI machine learning repository [20], [21], and [22]. The datasets WDBC, WPBC and SPECT are collected from the UCI machine learning repository.

The SBRCT dataset is collected from [22] and the other two datasets (prostate and colon) are collected from [21]. To demonstrate the superiority of our algorithm, we have selected both binary and multi-class datasets covering a range of biomedical applications, including cancer classification and disease prognosis. We also include datasets with a range of class distributions, spanning from slightly imbalanced (for example prostate) to highly imbalanced scenarios (for example spect). Table I shows the detailed characteristics of our dataset.

4.1.1 Dataset Characteristics

Table I: Dataset Characteristics

Dataset	Features	Samples	Classes	Class Distribution
Prostate [6]	6033	102	2	class 1: 59%, class 2: 49%
SBRCT [13]	2308	83	4	class 1: 34.9% , class 2: 13.3%, class 3: 21.7%, class 4: 30.1%
Colon [18]	2000	62	2	class 1: 35%, class 2: 65%
WDBC [6]	30	569	2	class 1: 62.8%, class 2:37.2%
WPBC [8]	33	198	2	class 1:65.5%, class 2:34.5%
SPECT [10]	44	267	2	class 1: 79%, class 2: 21%

4.2 Experimental Set up

We evaluated our proposed algorithm GA-SFS with two popular classification models K-Nearest Neighbor (KNN) and Support Vector Machine (SVM) [4], [11]. Since biomedical datasets are typically highly imbalanced, we use stratified k-fold cross-validation with k=5 to preserve the original class distribution in each fold, ensuring a more accurate and reliable evaluation [9], [16]. We use the following parameter setup for our Genetic Algorithm: number of generations: 15, population size=7, number of parents mating=7, tournament size=3, base mutation ate=0.05. To evaluate our model's performance, we use three metrics: accuracy, F1 score, and balanced accuracy. The reason for choosing the F1 score and balanced accuracy is that biomedical datasets are often imbalanced, so we want to ensure that all classes are correctly classified without favoring the majority class.

The Equation for F1-Score:

$$F1 = 2\times\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \text{where:} \quad \text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN} \qquad \text{Eq(12)}$$

The equation for Balanced Accuracy:

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right) \qquad \text{Eq(13)}$$

The equations for Accuracy:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \qquad \text{Eq(14)}$$

4.3 Results: With and Without Feature Selection

We present an experiment demonstrating that selecting relevant features significantly enhances model performance. Table II highlights the results, showcasing performance improvements across two different evaluation classifiers: KNN and SVM. The number of selected features is substantially reduced across six datasets, with reductions reaching nearly 50% or more in the Prostate, SBRCT, and Colon datasets. However, the original feature size for WDBC, WPBC, and SPECT is relatively smaller compared to the other three datasets. Despite this, the reduction in the number of features remains significant. In the case of KNN evaluation, our algorithm outperforms across all datasets when compared with no feature selection. There is a noticeable improvement in KNN performance with feature selection, particularly in the Prostate dataset (from 0.75 to 0.92), the SBRCT dataset (0.81 to 0.97), and the SPECT dataset (from 0.74 to 0.95). For SVM evaluation, our algorithm shows superior results in 6 out of 6 datasets. However, for the SPECT dataset, the changes are minimal. One possible reason could be the SVM algorithm itself is robust to redundant features. In both KNN and SVM evaluations, both large and small datasets show significant performance improvements with our feature selection approach.

Table II: Model Performance: With and Without Feature Selection

Dataset	Original Features	Selected Features	KNN		SVM	
			Without FS	FS(GA-SFS)	Without FS	FS(GA-SFS)
Prostate	6033	2948	0.75	0.92	0.87	0.98
SBRCT	2308	1144	0.81	0.97	1.00	0.96
Colon	2000	996	0.76	0.82	0.74	0.90
WDBC	30	13	0.92	0.97	0.90	0.98
WPBC	33	16	0.92	0.97	0,90	0.98
SPECT	44	22	0.74	0.95	0.95	0.99

4.4 GA-SFS vs. State of the Art Algorithms

Table III , Table IV , and Table V compares GA-SFS against Fast SFFS, demonstrating consistent improvements in accuracy, F1 score, and balanced accuracy across the six mentioned biomedical datasets. We compare our algorithm with the state-of-the-art Fast SFFS algorithm [16]. The Fast SFFS algorithm combines filter and wrapper methods for feature selection. It randomly selects an initial subset of features, typically around 20-25% of the total, which yields a higher AUC than using all the features. The candidate subset generated by the filter approach is then evaluated using k-fold cross-validation with a support vector machine (SVM), where a user-defined classification margin is applied as a wrapper [18].

Table III: GA-SFS vs State-of-the-art algorithm (Accuracy)

Dataset	KNN		SVM	
	GA-SFS	State-of-the-art Algorithm	GA-SFS	State-of-the-art Algorithm
Prostate	0.92	0.87	0.98	0.84
SBRCT	0.97	0.83	1.00	0.89
Colon	0.82	0.62	0.90	0.67
WDBC	0.97	0.96	0.98	0.90
WPBC	0.97	0.95	0.98	0.90
SPECT	0.95	0.77	0.99	0.92

Table IV: GA-SFS vs State-of-the-art algorithm (F1-Score)

Dataset	KNN		SVM	
	GA-SFS	State-of-the-art Algorithm	GA-SFS	State-of-the-art Algorithm
Prostate	0.92	0.87	0.98	0.84
SBRCT	0.96	0.85	1.00	0.89
Colon	0.80	0.62	0.90	0.67
WDBC	0.97	0.94	0.98	0.96
WPBC	0.97	0.94	0.98	0.94
SPECT	0.95	0.64	0.98	0.64

Table V: GA-SFS vs State-of-the-art algorithm (Balanced Accuracy)

Dataset	KNN		SVM	
	GA-SFS	State-of-the-art Algorithm	GA-SFS	State-of-the-art Algorithm
Prostate	0.92	0.87	0.98	0.84
SBRCT	0.94	0.86	1.00	0.89
Colon	0.76	0.68	0.88	0.72
WDBC	0.96	0.94	0.97	0.96
WPBC	0.96	0.95	0.98	0.95
SPECT	0.92	0.64	0.98	0.64

4.5 Efficiency and Practical Applications

The proposed algorithm's time complexity is influenced by chromosome refinement steps and fitness evaluation function, resulting in a time complexity of $O(G \cdot P \cdot F + K)$, where G is the number of generations, P is the population size, F is the number of features, and K is the complexity of model evaluation. The GA-SFS hybrid algorithm has real-world potential, especially in scenarios like early cancer detection [13], [22]. For instance, when working with gene expression data, which often contains thousands of features, the algorithm can identify the most critical genetic markers linked to specific cancer types. It will help narrow down the data to the most relevant features, it enables researchers and clinicians to build more accurate predictive models for identifying cancer at an early stage. This targeted approach not only helps diagnostics but also reduces computational complexity, making it practical for real-world biomedical applications.

Chapter 5:

Conclusion

In this study, we introduced GA-SFS, a hybrid feature selection algorithm that combines Genetic Algorithm (GA) and Sequential Forward Selection (SFS) to improve feature selection in high-dimensional biomedical datasets [3], [6]. By leveraging GA's global optimization capabilities and SFS's local refinement efficiency, GA-SFS reduces the redundant features effectively while also maintaining or improving classification accuracy. Our experimental results show that GA-SFS outperformed traditional feature selection methods, including Fast SFFS, across multiple biomedical datasets, achieving higher accuracy, balanced accuracy, and F1-Scores. Despite its advantages, GA-SFS has a high computational cost, primarily due to the iterative nature of the feature selection process of SFS. To address this, future research will focus on optimizing the algorithm's efficiency by integrating an advanced version of SFS and introducing adaptive tournament selection to be able to enhance Ga's selection process. Expanding GA-SFS to support unsupervised feature selection will widen the applicability of to a wider range of machine learning tasks in biomedical data analysis. This research will contribute to the field by providing an effective, hybrid feature selection approach that improves both model performance and interpretability, in various types of datasets such as disease classification and biomarker discovery.

5.1 Summary

This study presented a hybrid GA-SFFS approach for feature selection in high-dimensional biomedical datasets. In our two-stage hybrid method, we utilized the genetic algorithm and the sequential forward selection to select the relevant feature. The proposed method demonstrated

superior performance over an advanced SFFS-based state-of-the-art method across multiple metrics, including accuracy, balanced accuracy, and F1 score. A limitation of our algorithm is its high computational cost, as it adds one feature at a time during the Sequential Feature Selection (SFS) process [8], [16].

5.2 Limitations

A limitation of our algorithm is its high computational cost, as it adds one feature at a time during the Sequential Feature Selection (SFS) process. The iterative feature selection approach, while effective in refining the feature subset, will often result in increased processing time when dealing with large-scale biomedical datasets with thousands of features [9], [19]. Additionally, if there is a scenario in supervised learning where labeled data is scarce the adaptability of the model may decrease. Another challenge is the sensitivity to parameter tuning, specifically in the genetic algorithm stage, where mutation rate, crossover probability, and selection process can significantly impact the performance. Future improvements can have a focus on optimization in computational efficiency while mating a high performance rate using accuracy and robustness of the feature selection process.

5.3 Future Work

In future work, we plan to incorporate an advanced version of SFS and introduce adaptive tournament selection within the genetic algorithm to reduce time complexity. By adjusting feature selection criteria to be dynamic, we aim to make the algorithm more adaptable to various datasets. Additionally, we aim to adapt the algorithm for unsupervised feature selection, expanding its applicability to a broader range of tasks. Specifically, where labeled data is unavailable, making it more versatile for broader machine learning applications. Another avenue

for improvement is exploring parallel computing techniques to accelerate the GA-SFS execution, making it feasible for real time or large scale datasets. Lastly, The integration of deep learning based feature selection techniques, such as autoencoders, could enhance the models ability to capture complex relationships between features, improving final model performance.

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